

- 11 **B** **A** is wrong because magnitude of gravitational force GMm/r^2 decreases with increasing distance.

In an orbit, gravitational force provides centripetal force, and since centripetal force decreases, so is the centripetal acceleration (Thus **D** is wrong).

From Kepler's Third Law,

$$T^2 \propto r^3$$

T increases as r increases.

Hence angular velocity $\omega = 2\pi / T$ decreases with increasing distance (Thus **C** is wrong).

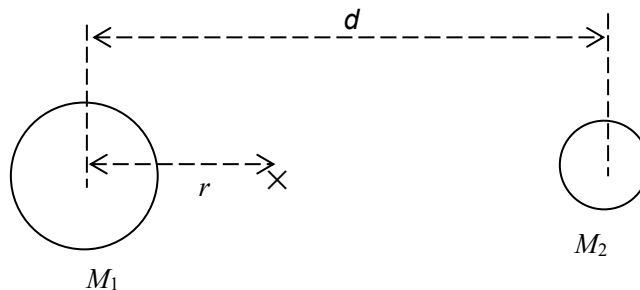
B is correct, as gravitational potential energy $-GMm/r$ is a scalar and it

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becomes less negative (hence increases) with increasing distance.

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D



Let the distance from M_1 be r from where it experiences no resultant gravitational force.

At r ,

$$\frac{GM_1}{r^2} = \frac{GM_2}{(d-r)^2}$$

Simplifying,

$$r = d \left(\frac{\sqrt{M_1}}{\sqrt{M_1} + \sqrt{M_2}} \right)$$

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A

A is incorrect as the amount of energy absorbed is equal to the latent heat